Causal KV Communication Between In-Context Operator Learners

Learning coupled dynamics from private local examples

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Can local examples determine a coupled trajectory?

Each agent knows examples of its own subsystem.
The new prediction depends on both subsystems.

$$\dot{x} = -ax + \kappa y, \qquad \dot{y} = -by + \lambda x, \qquad t \in [0, 1].$$

What each agent observes

Four local forcing–response examples and a new local initial value. The coefficients are hidden.

What they must predict

The pair of trajectories (x^*, y^*) for the new (x_0, y_0), using three exchanges of learned features.

Training: $a, b \sim U(0.6, 1.4)$, $\kappa, \lambda \sim U(-0.25, 0.25)$, and $x_0, y_0 \sim U(-1, 1)$.

A local operator maps forcing to response

For a prescribed input u, agent 1's equation has an explicit solution:

$$\mathcal{S}_1^{a,\kappa}(x_0, u)(t) = e^{-at}x_0 + \kappa \int_0^t e^{-a(t-s)}u(s)\,ds.$$

Its examples reveal this response rule. Agent 2 has the analogous operator $\mathcal{S}_2^{b,\lambda}(y_0, v)$.

Both maps take an initial value and a continuous forcing to a differentiable response: $\mathbb{R} \times C([0,1]) \to C^1([0,1])$.

The coupled problem closes the two inputs

Now each subsystem is forced by the other subsystem's unknown solution:

$$x^* = \mathcal{S}_1^{a,\kappa}(x_0, y^*), \qquad y^* = \mathcal{S}_2^{b,\lambda}(y_0, x^*).$$

These identities define the target; the networks exchange latent features to approximate it.

What is learned from examples?

The physical map

For fixed coefficients $\theta = (a, b, \kappa, \lambda)$, the ODE sends a new initial pair to a pair of functions:

$$\mathcal{G}_\theta : (x_0, y_0) \longmapsto (x^*(\cdot), y^*(\cdot)).$$

The distributed approximation

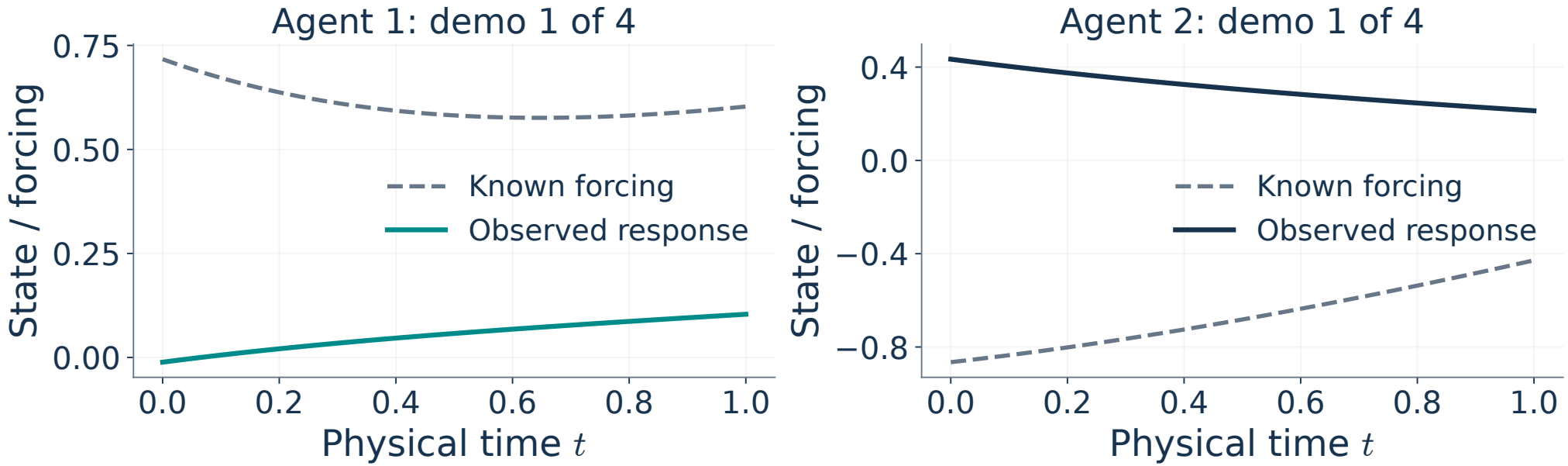
The networks read private Contexts $\mathcal{C}_1, \mathcal{C}_2$ and exchange Query features:

$$\mathcal{F}_\Theta^{(3)} : (\mathcal{C}_1, \mathcal{C}_2, x_0, y_0) \longmapsto (\widehat{x}, \widehat{y}) \in \mathbb{R}^{64\times 2}.$$

The weights Θ are trained to predict trajectories. The local operator is identified implicitly from examples; there is no explicit estimate of θ.

Predictions are sampled at $t_k = k/63$, $k = 0, \ldots, 63$.

One episode: examples first, then a new Query



Observed Context

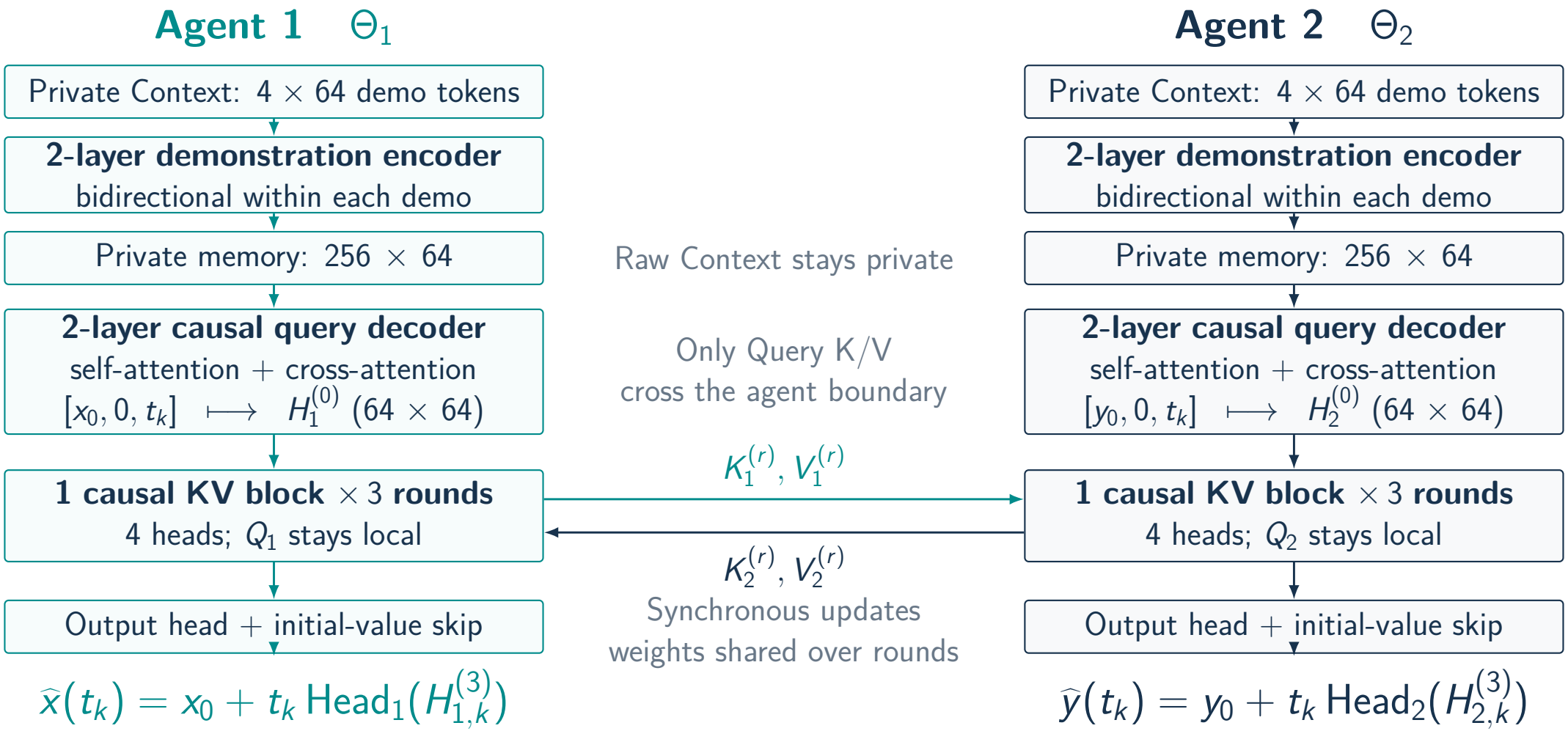
Four examples share each agent's hidden local parameters. The forcings u_m, v_m are known; the responses x_m, y_m are observed.

New coupled Query

Only the initial values change. The required forcings $u = y^*$ and $v = x^*$ are unknown and are not supplied.

One of four demonstrations per agent is shown, from ID episode 720.

Two independent ICON-inspired agents



From observed examples to Query features

Encode what is already known

Each demonstration token contains an initial value, a forcing, a response and time:

$$p_{i,m,k} = [z^{\mathrm{demo}}_{i,0,m},\ w_{i,m}(t_k),\ z_{i,m}(t_k),\ t_k].$$

Two **bidirectional** encoder layers process each complete example.

The four encoded sequences form private memory $\mathrm{Mem}_i \in \mathbb{R}^{256\times 64}$.

Here z_i is x or y, and w_i is the corresponding forcing u or v.

Initialize what must be predicted

The new Query contains a local initial value, a zero forcing placeholder and time:

$$q_{i,k} = [z_{i,0},\ 0,\ t_k].$$

Two **causal** decoder layers attend to earlier Query times and to the complete private memory:

$$H_i^{(0)} = \mathrm{Decoder}_i(\mathrm{Mem}_i, q_i) \in \mathbb{R}^{64\times 64}.$$

The zero means that the coupled forcing is unknown.

What travels between the two agents?

An agent sends features of its current Query, not a decoded trajectory.

For each head h, project the normalized state at round r:

$$[Q_{i,h}^{(r)}\ K_{i,h}^{(r)}\ V_{i,h}^{(r)}] = \mathrm{LN}_i(H_i^{(r)})W_{QKV,i,h} + b_{QKV,i,h}.$$

Keep the query Q local

It determines which features this receiver reads. The raw Context and private memory remain with their agent.

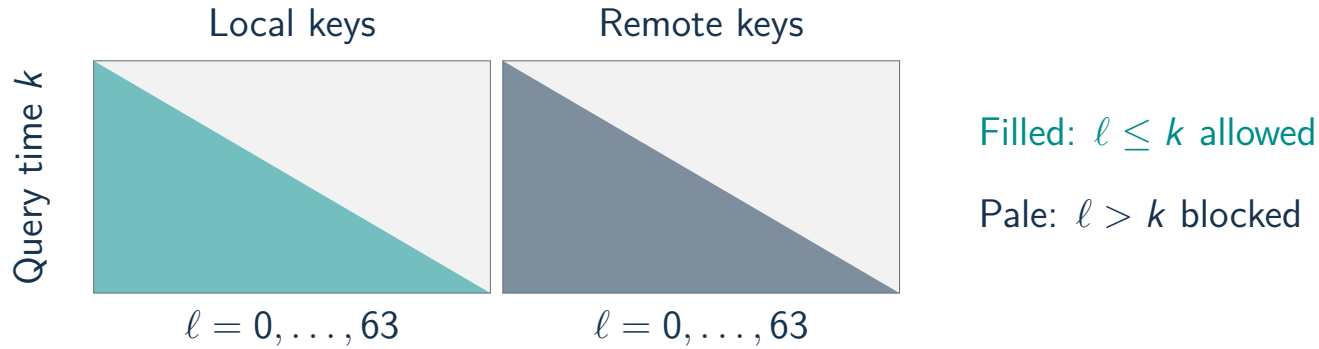
Send keys K and values V

Keys determine relevance; values carry the features being averaged. Only these two tensors cross between agents.

Four heads of width 16: each head is 64×16; each transmitted K or V tensor has shape $[4, 64, 16]$.

Read both agents at the same physical time

At Query time t_k, either agent may read local and remote features only at $t_\ell \le t_k$.



Join the two banks, $\overline{K}_i = [K_i; K_j]$ and $\overline{V}_i = [V_i; V_j]$. For one head, attention is

$$A_i = \mathrm{softmax}_{\mathrm{row}}\left(\frac{Q_i \overline{K}_i^\top}{\sqrt{16}} + \underbrace{[\,M \;\; M\,]}_{64\times 128} + [\,0 \;\; b_{\mathrm{remote},i,h}\,]\right)\overline{V}_i.$$

$M_{k\ell} = 0$ for $\ell \le k$ and $-\infty$ otherwise; the learned remote bias is per head. Round/head indices are suppressed.

The same time mask is applied to both Query banks. Fully observed demonstration memory remains available through the private decoder.

Use the change caused by remote information

Let $A_{i,\mathrm{local}}$ be the same causal attention with only agent i's own keys and values. Concatenate heads before the output projection.

First, correct the hidden state

$$U_i = H_i^{(r)} + g_i(A_i - A_{i,\mathrm{local}})W_{O,i}, \qquad g_i = \sigma(\gamma_i).$$

The learned scalar gate scales the difference between the two readings.

Then, apply the corresponding feed-forward correction

$$H_i^{(r+1)} = U_i + g_i \left[\mathrm{FFN}_i(\mathrm{LN}_{F,i}(U_i)) - \mathrm{FFN}_i(\mathrm{LN}_{F,i}(H_i^{(r)}))\right].$$

If all remote slots are masked, the two attentions coincide:

$$A_i = A_{i,\mathrm{local}} \quad \Longrightarrow \quad H_i^{(r+1)} = H_i^{(r)}.$$

Three exchanges, followed by physical readout

Both agents form their outgoing packets from the old states before either updates. The same communication block is reused within each agent:

$$H^{(0)} \xrightarrow{\Phi_\Theta} H^{(1)} \xrightarrow{\Phi_\Theta} H^{(2)} \xrightarrow{\Phi_\Theta} H^{(3)}.$$

Here $H^{(r)} = (H_1^{(r)}, H_2^{(r)})$; the agents have independent weights.

Decode trajectories after the last round

$$\widehat{x}(t_k) = x_0 + t_k \,\mathrm{Head}_1(H_{1,k}^{(3)}), \qquad \widehat{y}(t_k) = y_0 + t_k \,\mathrm{Head}_2(H_{2,k}^{(3)}).$$

Multiplication by t_k enforces the initial conditions exactly. The decoded trajectories are not fed back as messages.

Three rounds are three latent corrections, not three physical time steps or a proved convergent iteration.

Train communication through the coupled trajectories

The coupled branch compares both predictions after all three rounds with the exact ODE solution:

$$\mathcal{L}_{\text{coupled}} = \frac{1}{BN} \sum_{e=1}^{B} \sum_{k=0}^{N-1} \left[(\widehat{x}_e(t_k) - x_e^*(t_k))^2 + (\widehat{y}_e(t_k) - y_e^*(t_k))^2 \right].$$

The two state errors are summed and averaged over $B = 32$ episodes and $N = 64$ times.

Where the targets come from

The data generator solves the hidden linear system:

$$\begin{pmatrix} x_e^*(t_k) \\ y_e^*(t_k) \end{pmatrix} = e^{t_k B_{\theta_e}} \begin{pmatrix} x_{0,e} \\ y_{0,e} \end{pmatrix}, \quad B_\theta = \begin{bmatrix} -a & \kappa \\ \lambda & -b \end{bmatrix}.$$

Coefficients and targets enter the loss, not the coupled Query.

Why both agents learn

Gradients pass through all three exchanges. An error in agent 1's output can update agent 2 through its outgoing packet:

$$\mathcal{L}_x \longrightarrow (K_2, V_2) \longrightarrow \Theta_2.$$

Keep the local response maps supervised

A separate local task supplies a known forcing, its own Context and a new initial value. The same encoder, decoder and head predict the forced response without communication:

$$\mathcal{L}_{\mathrm{local},i} = \frac{1}{BN} \sum_{e=1}^{B} \sum_{k=0}^{N-1} \left(\hat{z}_{i,e}^{\mathrm{loc}}(t_k) - z_{i,e}^{*,\mathrm{loc}}(t_k) \right)^2, \qquad i = 1, 2.$$

This auxiliary supervision helps each backbone learn its local operator.

One joint objective

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{coupled}} + 0.2\left(\mathcal{L}_{\mathrm{local},1} + \mathcal{L}_{\mathrm{local},2} \right).$$

The known forcing belongs only to the independent local tasks. The coupled branch still has unknown forcing; no coefficient-regression or intermediate-round loss is used.

Training once, predicting new episodes

Joint training from scratch

Both backbones and communication blocks are trained for 50,000 AdamW updates, with batch size 32 and depth $R = 3$.

Validation selects the final model. The experiment uses one initialization seed.

Frozen-weight evaluation

Test on 1,024 ID and 1,024 OOD episodes.

Each new Context is read without updating weights. Only Contexts and local initial values are supplied; the Query forcing remains unknown.

We measure the relative error of the whole pair of sampled trajectories:

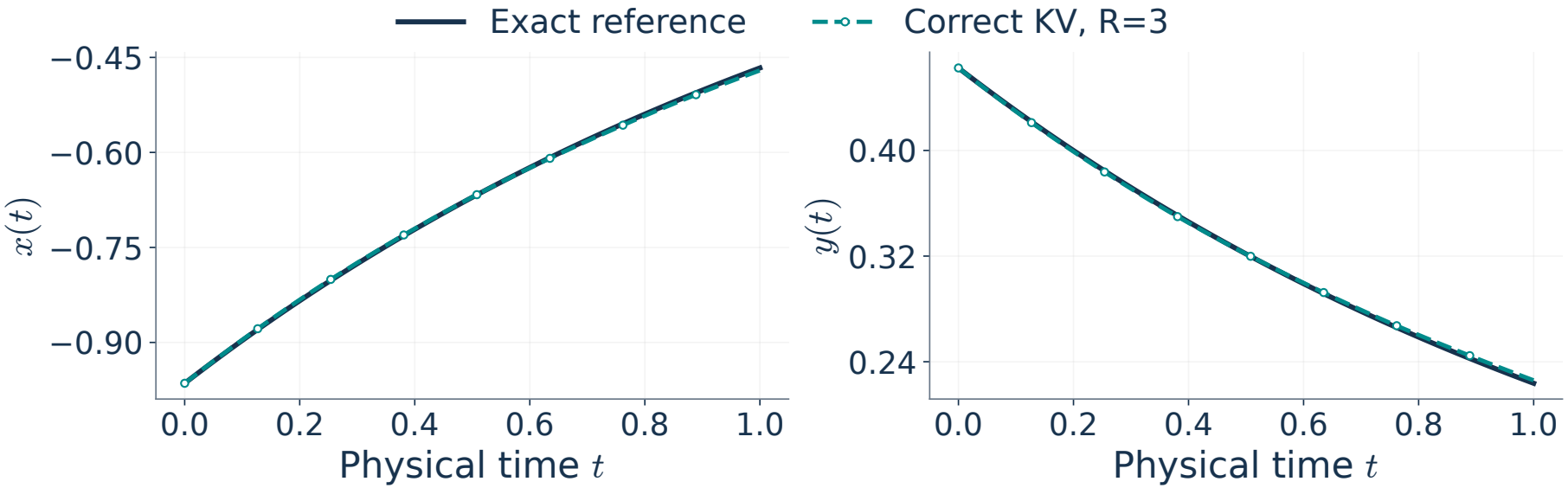
$$\mathcal{E}_e = \frac{\|[\widehat{x}_e, \widehat{y}_e] - [x_e^*, y_e^*]\|_F}{\max(10^{-8}, \|[x_e^*, y_e^*]\|_F)}, \qquad \overline{\mathcal{E}} = \frac{1}{1024} \sum_e \mathcal{E}_e.$$

Test episodes are not used for model selection.

In-distribution prediction

ID distribution: $a, b \sim U(0.6, 1.4)$; $\kappa, \lambda \sim U(-0.25, 0.25)$.

Initial values: $x_0, y_0 \sim U(-1, 1)$, as in training.

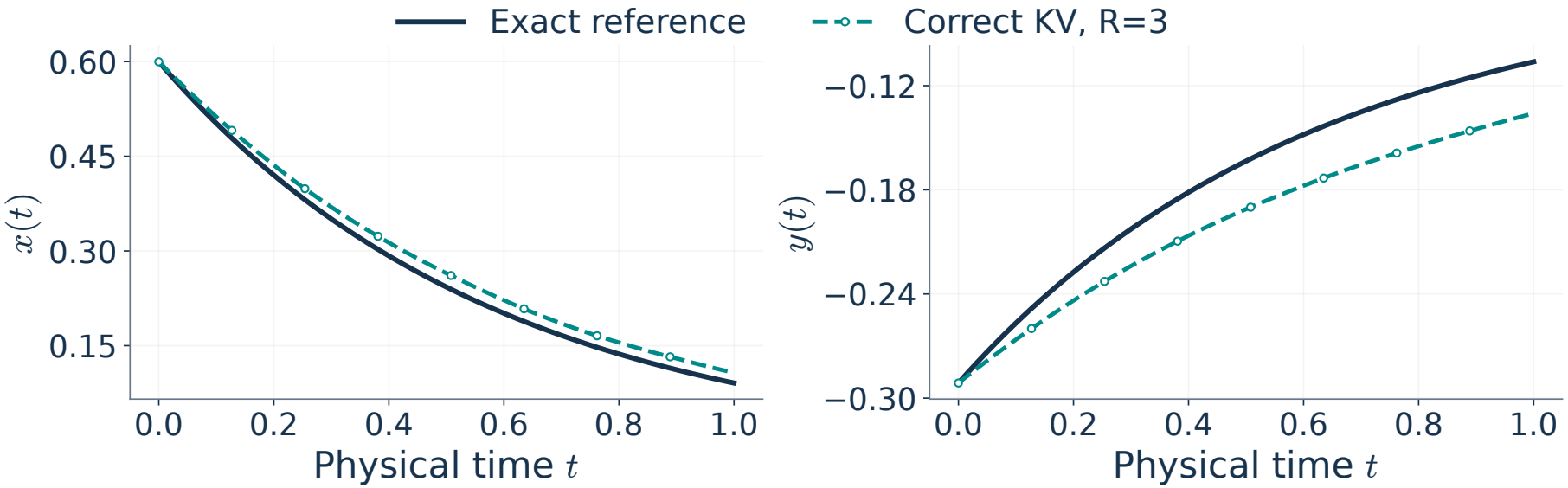


Shown case: ID episode 720 (rounded)
$(a, b, \kappa, \lambda) \approx (0.6092, 1.1017, 0.2437, -0.1800)$,
$(x_0, y_0) \approx (-0.9645, 0.4625)$.

Correct KV, $R = 3$
Case: 0.2841%; mean: 0.3571%.
Median-ranked; 1,024 episodes.

Out-of-distribution prediction

OOD decays: each of a, b is drawn equally from $U(0.25, 0.5)$ or $U(1.5, 1.9)$.
$|\kappa|, |\lambda| \sim U(0.3, 0.55)$ with independent equiprobable signs; $x_0, y_0 \sim U(-1, 1)$.



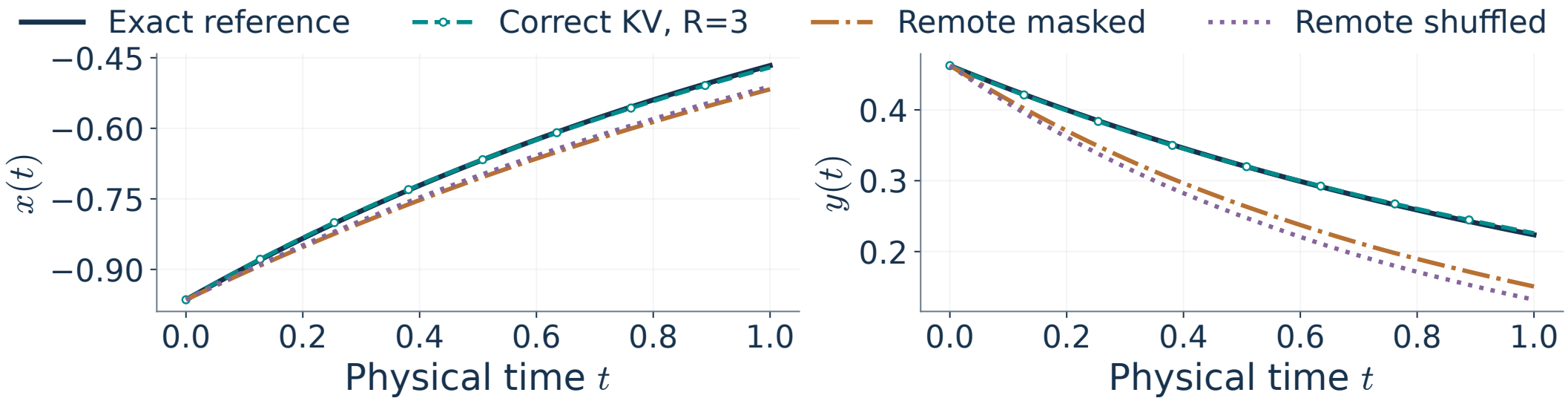
Shown case: OOD episode 349 (rounded)
$(a, b, \kappa, \lambda) \approx (1.5177, 0.3051, 0.5126, 0.4767)$,
$(x_0, y_0) \approx (0.5997, -0.2913)$.

Correct KV, $R = 3$
Case: 8.4756%; mean: 9.7181%.
Median-ranked; 1,024 episodes.

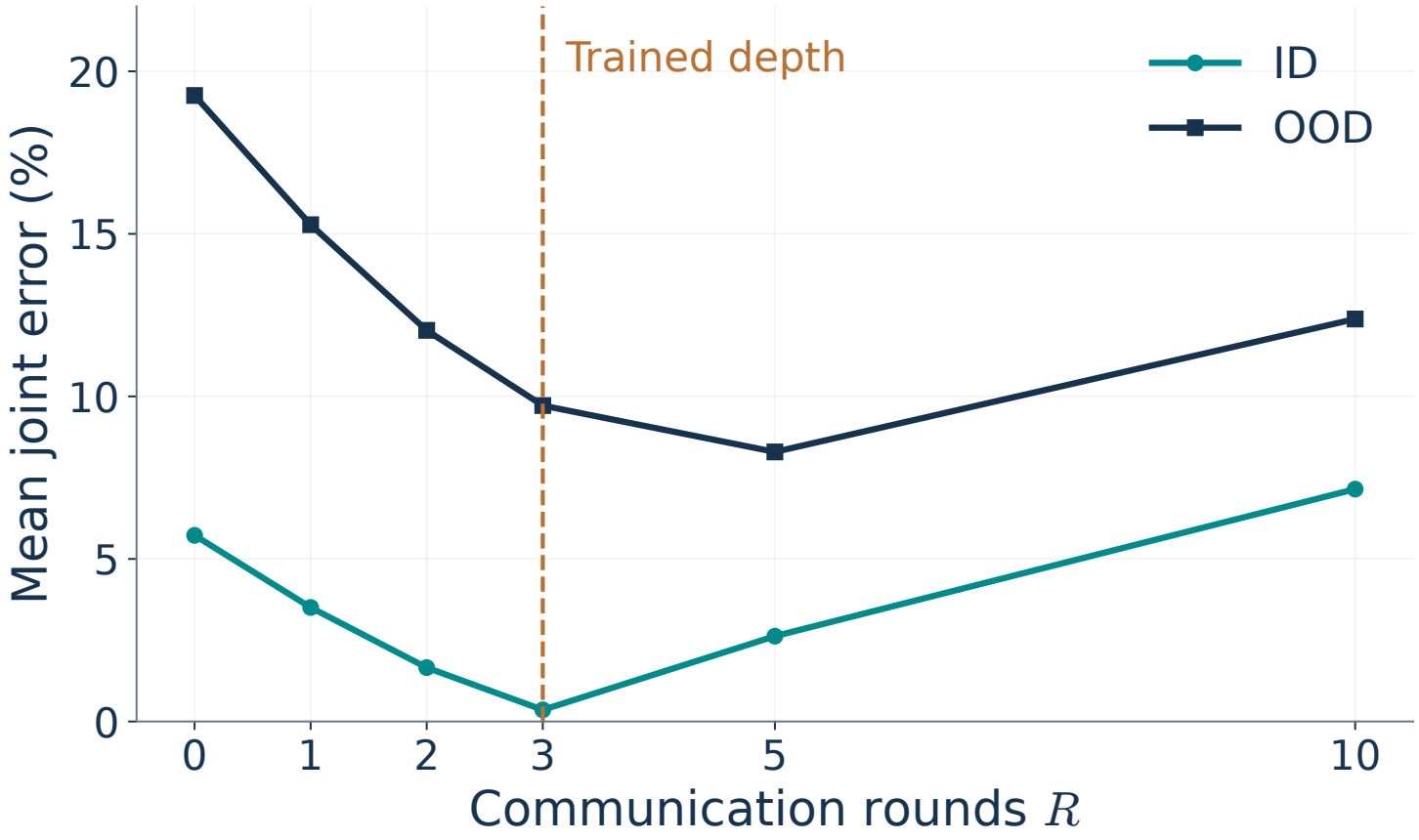
Does remote KV matter?

$R = 3$ intervention	ID mean	OOD mean
Correct KV	0.3571%	9.7181%
Remote masked	5.7257%	19.2560%
Remote shuffled	9.3305%	24.1632%

- Same model; 1,024 episodes per split. Plot: ID 720, with the parameters shown on slide 15.
- Masked: all remote slots blocked.
- Shuffled: roll episodes within batches of 32 at each round; preserve time order.



Depth interventions on the same model



- Same checkpoint; only $R = 3$ was trained.
- ID mean: 0.3571% at $R = 3$, 7.1479% at $R = 10$.
- More rounds do not imply convergence.

Conclusions and the next question

- Private local observations and three causal KV exchanges recover weakly coupled trajectories in distribution.
- Masking and shuffling support the use of episode-specific remote information.
- Reliability at untrained depths, OOD/strong coupling and communication efficiency remain open.

Research direction

Characterize how local approximation error and causal communication sensitivity control physical output error.

Thank You

Questions and Discussion

A1. Causal prefix dependence

$$\Pi_{\le k} H_i^{(r+1)} = \Psi_{i,k}\big(\Pi_{\le k} H_1^{(r)}, \Pi_{\le k} H_2^{(r)}\big).$$

- Fix the observed Contexts; $\Pi_{\le k}$ selects Query rows $0, \dots, k$.
- The causal decoder and both KV banks read only the Query prefix; all other operations are tokenwise.
- Induction preserves this property across rounds; complete demonstration histories remain available.
- Future-perturbation audits found earlier-time change 0, tolerance 2×10^{-5}, through round 10.

A2. Local communication sensitivity

$$J(H) = D_H \Phi_\Theta(H), \qquad \hat{s}_{\max} \approx \sqrt{\lambda_{\max}(J^\top J)}, \qquad H \in \mathbb{R}^{8192}.$$

$\lvert\kappa\rvert = \lvert\lambda\rvert$	0	0.25	0.8	1.8
Mean gain estimate at $R = 3$	1.0014	1.0432	1.1381	1.2666

- Two episodes per strength; 16 power iterations with Jacobian-vector products.
- Singular gain measures local Euclidean amplification, not spectral radius.
- These finite estimates provide no uniform contraction or convergence certificate.

A3. LayerNorm neutral directions

For one scalar shift per time row, $D_i = c_i \mathbf{1}_{64}^\top$ with $c_i \in \mathbb{R}^{64}$:

$$\mathrm{LN}_i(H_i + D_i) = \mathrm{LN}_i(H_i), \qquad \Phi_\Theta(H + D) = \Phi_\Theta(H) + D.$$

$$J(H)D = D, \qquad \|\Phi_\Theta(H + D) - \Phi_\Theta(H)\|_F = \|D\|_F.$$

- Normalized attention/FFN terms ignore the shift; residual connections carry it unchanged.
- At differentiable states, the shift is a unit-gain direction: no strict full-state Euclidean contraction.
- The LayerNorm output head also ignores it; physical-output stability is a separate question.

A4. Strong coupling and OOD tails

Strong-coupling sweep

$\lvert\kappa\rvert = \lvert\lambda\rvert$	Mean error
0	0.2375%
0.125	0.3138%
0.25	0.5558%
0.5	7.2601%
0.8	20.1786%
1.2	37.3025%
1.8	61.9308%

- 128 episodes per strength, $R = 3$; ID decay and initial-state ranges.
- Training has $|\kappa|, |\lambda| \leq 0.25$.

OOD at the trained depth

Mean	9.7181%
99th percentile	26.0917%
Maximum	244.0816%
Above 20% error	49/1,024

- One initialization seed; episode variation is not cross-seed uncertainty.
- Both the local operators and the coupled task shift outside training.

A5. Causal-KV communication payload

$$\underbrace{2}_{\text{senders}} \times \underbrace{2}_{K,V} \times \underbrace{4 \cdot 64 \cdot 16}_{\text{tensor entries}} \times \underbrace{4}_{\text{bytes}} = 65{,}536 \text{ bytes/round}.$$

- Each sender transmits float32 K, V with shape $[4, 64, 16]$.
- Total payload: **64 KiB per round**, or **192 KiB per episode** at $R = 3$.
- Logical tensor sizes from single-GPU execution; no compression or network-latency measurement.

A6. Extending the same model to 1.5 seconds

Direct extension

$$\mathcal{T}_{\text{ext}} = \{k/63 : k = 0, \ldots, 94\} \cup \{1.5\}.$$

- Keep weights and Context fixed on $[0, 1]$; extend only the Query grid to 96 times.
- Run three KV rounds; preserve the first 64 times and use final step $1/126$.

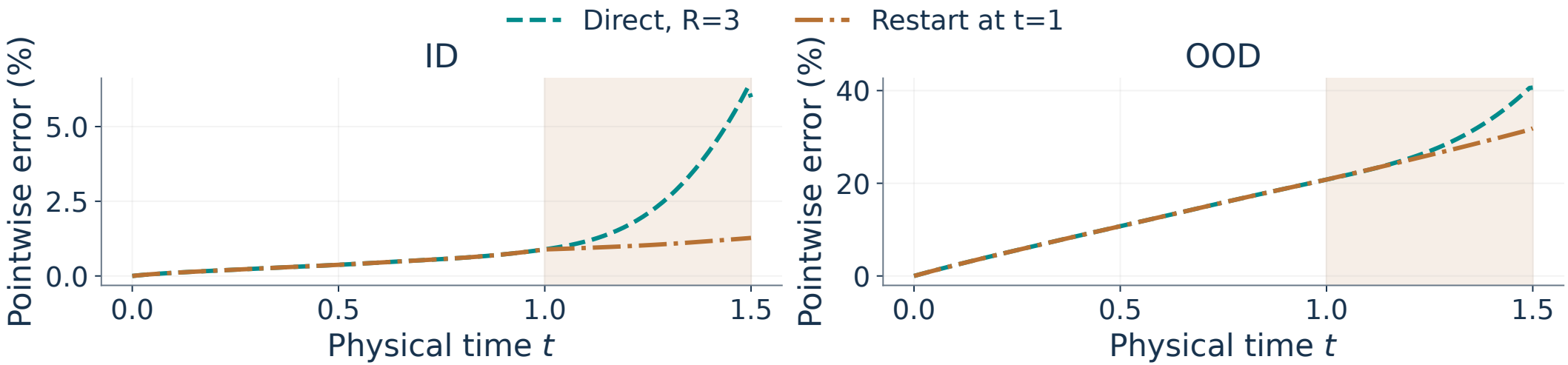
Predicted-endpoint restart

$$(\widetilde{x}_0, \widetilde{y}_0) = (\widehat{x}(1), \widehat{y}(1)).$$

- Own predicted endpoint; reuse memory and reset hidden states.
- $\tau = t - 1 \in [0, 0.5]$: 33 times, then three new rounds.

Both protocols use the same matrix-exponential reference; seconds denote benchmark time units.

A7. Error growth through $t = 1.5$



Mean pointwise relative error over 1,024 episodes per split; shading marks $t > 1$.

Split	Method	$[0, 1.5]$	$(1, 1.5]$	$t = 1.5$
ID	Direct	0.89%	2.81%	5.98%
ID	Restart	0.47%	1.05%	1.27%
OOD	Direct	13.33%	28.07%	40.65%
OOD	Restart	13.00%	25.84%	31.86%

The table uses joint interval errors; these are not averages of the plotted curve.

Tail absolute RMS (direct / restart):
ID 0.00368/0.00150; OOD 0.05886/0.05883.

Restart improves ID continuation; OOD errors remain large.